



Micro scale 3D FEM simulation on thermal evolution within the porous structure in selective laser sintering

F.R. Liu*, Q. Zhang, W.P. Zhou, J.J. Zhao, J.M. Chen

Institute of Laser Engineering, Beijing University of Technology, 100124 Beijing, China

ARTICLE INFO

Article history:

Received 20 January 2012

Received in revised form 18 May 2012

Accepted 20 May 2012

Available online 27 May 2012

Keywords:

Finite element model (FEM)

Selective laser sintering (SLS)

Thermal evolution

Porous structure

ABSTRACT

A micro scale 3D finite element model (FEM) with consideration of powder arrangements was developed by selective laser sintering (SLS), where a multi-layer powder stacking model with cubic powders interlaced each other was established to describe the porous powder bed. All the powders directly exposed to laser irradiations were loaded with the Gaussian function of heat flux, and the non-linear thermal conductivity and specific heat owing to temperature change and phase transformation were considered. Comparison between the modeling and experiment indicated that both the simulated length of the sintered piece and shrinkage depth of the powder bed agreed well with the experiment data. The temperature field of laser sintering was intermittent in the micro scale due to the discretely distributed particles with maximum temperature produced in the top layer of the powder bed, and two primary ways of bonding were found in the powder bed during laser sintering.

© 2012 Elsevier B.V. All rights reserved.

1. Introduction

Selective laser sintering (SLS) is a technology to fabricate a three dimensional (3D) object layer by layer, which is first proposed by Deckard (1986). Since then, a lot of materials including polymer, metal, ceramic even compound powders have been employed to form the functional parts. Song et al. (2007) studied the manufacturing of phenol formaldehyde resin (PF resin)/silica sand composites and qualified patterns by SLS were obtained. Salmoria et al. (2009) fabricated the starch-cellulose and cellulose acetate scaffolds by SLS for tissue engineering, and the influence of process parameters on scaffold properties was addressed. Yan et al. (2011) investigated the carbon fibre/polyamide-12 composites by SLS also aiming at manufacturing high performance components.

Since the physical process of SLS associates with complex modes of heat and mass transfer, some theoretical models have been developed in the past years. Anestiev and Froyen (1999) proposed a theoretical model to describe the rearrangement process of the powder bed and analyzed the interaction between two particles during the SLS process. Zhang and Faghri (1999) built up an analytical model to theoretically investigate the melting behavior of a subcooled powder bed with mixed powders. The temperature distributions in the liquid and solid phases were respectively obtained by an exact solution and an integral approximate solution respectively. Gusarov et al. (2003) established an analytical model to

elucidate contact thermal conductivity of a powder bed in SLS, where the small contact necks were proposed to bond the particles. Chen and Zhang (2007) proposed an analytical model to study laser sintering of the two-component metal powders.

In order to clearly capture the transient variations of the temperature, stress or mass in the powder bed, some numerical models were also developed. Nelson et al. (1993) used 1D finite element method for the density prediction of sintered polycarbonate. Williams and Deckard (1998) developed a 2D model for laser sintering of amorphous polymers. However, it was regarded that the 1D or 2D modeling was not suitable to address the sintering because the significant temperature gradient existed at the edges of the part. Therefore, some 3D models were subsequently developed with consideration of improving the accuracy of predictions. Bugeda et al. (1999) established a numerical model to study the temperature and density distribution by a coupled model with the powder bed assumed as a compact box. The temperature dependent density change of the powder bed was first obtained by the analytical models, then coupled into the numerical simulation. Similarly, Dai and Shaw (2004) built up a 3D FEM model to study the thermo-mechanical behavior of laser sintering of metal and ceramic powders. Such properties as thermal conductivity, radiation and convection of the compact powder box were obtained by analytical models. Dai and Shaw (2005) further developed their model to study the effect of volume shrinkage during laser densification. A criterion was used to judge whether the element in the powder box remained as powder or became liquid during the simulation. Kolossov et al. (2004) established a 3D finite element model to study temperature evolution on the compact powder box using

* Corresponding author. Tel.: +86 010 67396552; fax: +86 010 67392773.
E-mail address: liufr@bjut.edu.cn (F.R. Liu).

the temperature dependent thermal conductivity and specific heat. More recently, Dong et al. (2009) developed a 3D finite element model to simulate the SLS process by a bi-level structure integration procedure with the powder bed also simplified as a compact box.

From the literature review, we know that most of the models developed, including analytical and numerical models, based on the assumption with the homogeneous powder bed, namely, the powder bed was assumed as a uniform compact box with the laser beam directly struck on the top surface of the box. Less attention has been paid on the thermal behavior of the local powder bed. The purpose of the present study is to develop a 3D finite element model (FEM) in a micro scale to investigate the characteristics of the temperature field within the powder bed in SLS.

2. Model description

2.1. Formation of the micro-scale model

In order to simulate thermal evolution within the porous structure during the SLS process, a three-dimensional (3D) multi-layer powder stacking model is built up with the assumptions summarized as follows:

- (i) The powder bed is made up of layers of powders.
- (ii) The particle is assumed cubic block and distributed uniform in each layer.
- (iii) The porosity of each layer is the same; therefore, the porosity of the powder bed can be predicted by any layer.
- (iv) Stacking way of the layers follows the method shown in Fig. 1(a), that is, each particle in the up layer is positioned on the center of the neighboring four particles of the bottom layer.
- (v) The propagation of the melting pool is not considered in the simulation.

For SLS, the partially or completely melting of particles occurs during the sintering process, as a result, the melting depth of the powder bed is usually less than the triple particle size as compared to another rapid prototyping technology, namely selective laser melting (SLM), where several particles are melted simultaneously for a higher laser power input. Therefore, a three-layer powder stacking model was constructed to simulate the thermal field produced by SLS in the present study. As shown in Fig. 1(c), the first layer of the powder bed is bottom layer with 4×4 block particles attached with the matrix. The second layer of the powder bed is called middle layer with 3×3 particles stacking on the bottom layer. Following the method depicted in (iv), the third layer is named as top layer which repeats the structure of the first layer. Then the length of the construction (L) is obtained by

$$L = 3s + D \quad (1)$$

where s is the distance between two neighboring particles, and D is particle diameter. A coefficient, R_a , is defined to characterize the ratio of the distance between two particles to particle diameter

$$R_a = \frac{s}{D} \quad (2)$$

Obviously, R_a is the function of porosity. According to the assumption (iii), the porosity of the powder bed can be represented by the top layer. Considering the continuity of the structure, porosity of the powder bed is calculated within the dash line shown in Fig. 1(a). By calculation, R_a is obtained ranging from 1.0 to 1.5 for the real sintering process. When there are no pores existing in the

powder bed, namely solid metal, R_a is equal to 1.0. While for the real compact powder bed, R_a is 1.4 corresponding to the porosity of 50%. In the present study, the porosity of 50% is used in the simulation in order to keep the real porosity of the compact powder bed.

In the paper, numerical simulation was executed by the ANSYS code. A 10-Node tetrahedral element was adopted for the thermal analysis. As shown in Fig. 1(c), a fine mesh was used in the scanning region in order to capture temperature distribution more accurately, while a coarse mesh was employed in remote areas to save the calculation time. The total number of the element in the model was about 60,000.

2.2. Model of the SLS process

The heat transfer behavior in the SLS process is described by the Fourier thermal equilibrium equation written as

$$\lambda \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) + \dot{q} = \rho C \dot{T} \quad (3)$$

where λ is thermal conductivity, ρ is density of the examined powder, C is specific heat and $\dot{q}$ is rate of heat generation by laser scanning.

Laser scanning is considered as a heat flux moving at a constant velocity from one side to another. The laser power follows the Gaussian distribution

$$I = \frac{2AP}{\pi r^2} \exp \left(-\frac{2((x - v \times t + r)^2 + y^2)}{r^2} \right) \quad (4)$$

where I is thermal flux density of the laser beam, P is laser output power, v is scanning speed, and r is the radius of laser beam, while A is absorption coefficient affected by laser wavelength and surface conditions as well as physical properties of the material. Absorption coefficient of 60% for the pure Fe powder under 1.06 μm wavelength irradiations came from Tolochko et al. (2000). All the powders directly exposed to laser irradiations were loaded with the Gaussian function of heat flux by Eq. (4). An ANSYS Parametric Design Language (APDL) was used to model the moving heat source.

Phase transformation of powders from solidus to liquid is involved in the SLS process, and latent heat of fusion (ΔH_{melt}) is considered by the change of enthalpy given by

$$H = \int_{T_0}^T \rho \times c dT \quad (T < T_m)$$

$$H = \int_{T_0}^T \rho \times c dT + \Delta H_{\text{melt}} \quad (T \geq T_m) \quad (5)$$

where H is enthalpy, T_0 is reference temperature which is set to 293 K in the paper, and T_m is melting point of the material.

In order to avoid the oxidation of the Fe powder under the elevated temperature, a shielding gas of Ar was employed during laser scanning in the experiment. Therefore, free heat convection is considered in the model:

$$q = h_f(T - T_0) \quad (6)$$

where h_f is heat transfer coefficient, and T_0 is absolute temperature of the surrounding Ar gas, which is set at 293 K in the present simulation. Radiation is also considered as follows:

$$q = \epsilon \Omega A_1 F_{12}(T^4 - T_0^4) \quad (7)$$

where ϵ is emissivity, Ω is Stephan–Boltzmann constant ($5.47 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$), A_1 is radiation area, F_{12} is form factor from

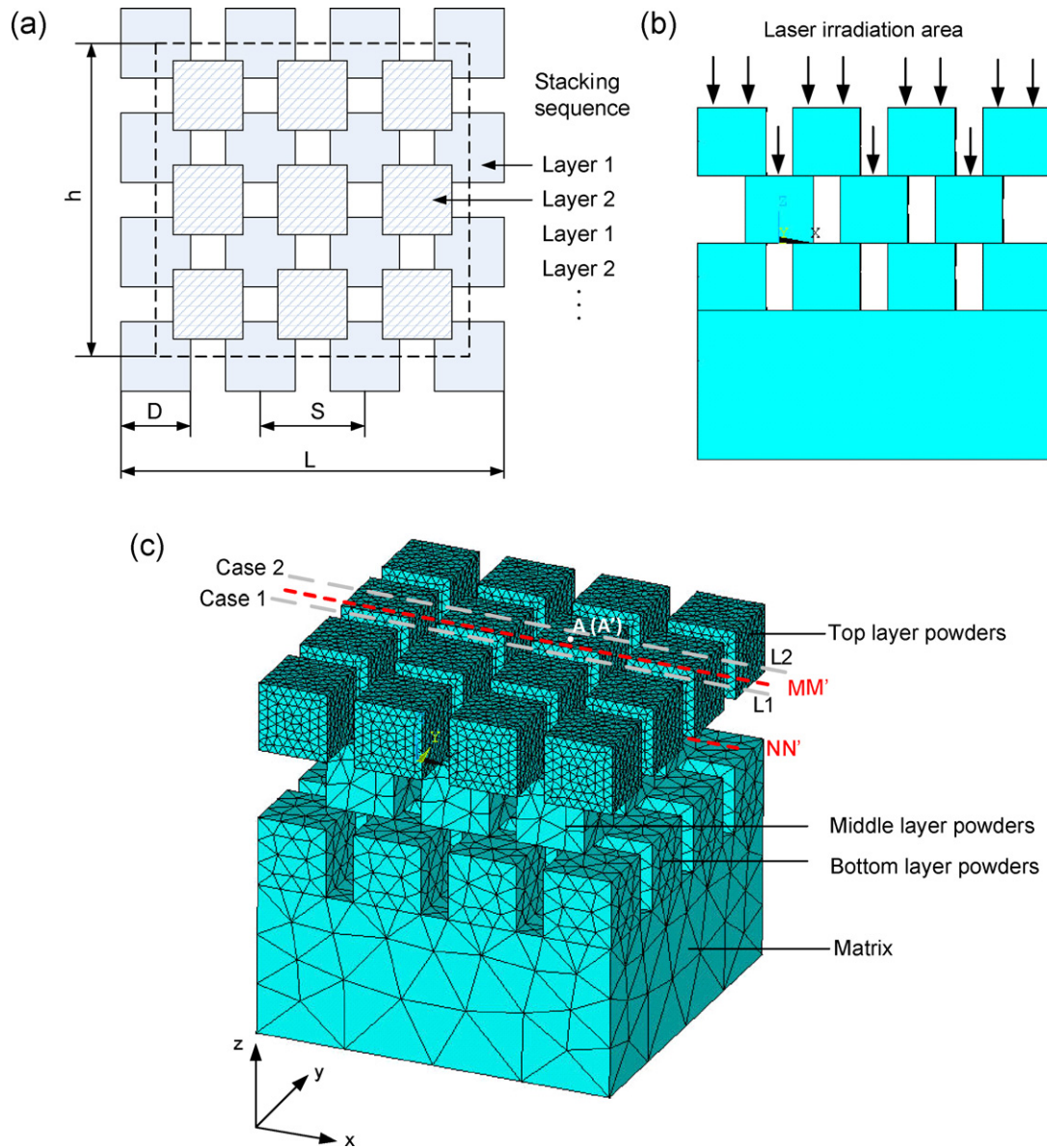


Fig. 1. Construction of the 3D multi-layer powder stacking model (a) the stacking strategy of the powders; (b) laser irradiation region; (c) 3D schematic for the simulation.

the radiation surface to surrounding shielding gas, and T is absolute temperature of the radiation surface.

2.3. Verification of the numerical model

In order to verify the numerical model, the length of the sintered piece and shrinkage depth of the powder bed were simulated and compared with the experimental results. From previous study of Dai and Shaw (2005) on the volume shrinkage in SLS, volume shrinkage was accomplished by the movement of the upper node of an element downward together with the shrinkage percentage equal to the porosity fraction of the powder bed. In a similar way, shrinkage depth is calculated by the melting depth ($\bar{t}_m$) of the powder bed as follows

$$\bar{t} = \sqrt[3]{p} \times \bar{t}_m \quad (8)$$

where $\bar{t}$ is the estimated shrinkage depth, p is porosity of the powder bed. The shrinkage of the different regions of the powder bed

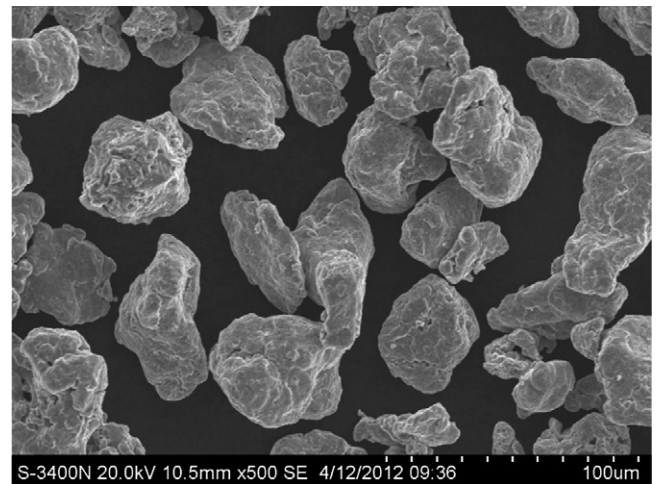


Fig. 2. Morphology of the as-received pure Fe powder.

Table 1
Material constants and initial conditions.

Material constants						Unit
Pure Fe	Thermal conductivity	73.3 (293 K)	48.6(673 K)	38.9 (873 K)	29.7 (1073 K)	W/m K
	Specific heat	444 (293 K)	523 (473 K)	611 (673 K)	791 (1073 K)	J/kg K
	Density	7800 (293 K)				kg/m ⁻³
	Latent heat	300 (1808 K)				kJ/kg
316 L	Thermal conductivity	14 (293 K)	18 (573 K)	21 (773 K)	29 (1373 K)	W/m K
	Specific heat	450 (293 K)	545 (573 K)	570 (773 K)	670 (1373 K)	J/kg K
	Density	7980 (293 K)				kg/m ⁻³
Initial conditions						Unit
Heat transfer coefficient		25				W/m ² K
Coefficient of emissivity		0.8				–
Absorption coefficient of the powder bed		60%				–
Initial temperature of the powder bed		293				K

varies in the micro scale, so, average melting depth ($\bar{t}_m$) of the powder bed is estimated by

$$\bar{t}_m = \frac{1}{N} \sum_{i=1}^N t_{mi} \quad (9)$$

where t_{mi} is the simulated melting depth of a small region. N is the number of the small region. Five typical small regions, namely, one top surface center and four surface corners of a powder in the top row, are adopted in the paper. Since shrinkage depth is affected by the process parameter and the porous structure of the powder bed in the micro scale, two kinds of typical laser scanning strategies were also studied in the paper. As shown in Fig. 1(c), for case 1, laser scans along the L1, that is, the central line between two rows of powders (the second- and third- row) in the top layer, while for case 2, laser moves along the central line of one row powders (the second row) in the top layer. The average shrinkage depth of the powder bed is further given with consideration of laser scanning strategies

$$t_{\text{simu}} = \frac{1}{2} \bar{t}_1 + \frac{1}{2} \bar{t}_2 \quad (10)$$

where t_{simu} stands for the simulated shrinkage depth, and $\bar{t}_1$ represents the average shrinkage depth of the sintered sample when laser scans along the L1, while $\bar{t}_2$ is that of the sample when laser scans along the L2. Therefore, 2×5 data in total were used to calculate the averaged shrinkage depth.

The length of the sintered piece was obtained by simulating the length of the melting pool along the direction of laser scanning. The simulated data are got by averaging the total data from two typical cases (case1 and case2). Each case also has two circumstances, that is, the incidence of laser beam is just on the top row and the second row separately.

2.4. Material properties

In the present study, the pure Fe powder was used to carry out the SLS process for verification. The pure Fe powder was provided by Beijing YouxingLian Nonferrous Metals Company Limited, made with electrolysis method. For this research, the provided Fe powder particles used for the experiments are irregular shaped, as shown in Fig. 2. The chemical composition of the powder (wt%) is: Mn 0.12, Si 0.1, C 0.08, S 0.06, P 0.02, and Fe balance. The melting point of the pure Fe powder is 1808 K, while the boiling point is 3023 K. The particles were measured by laser particle size analyzer (Mastersizer2000, Malvern Instrument Ltd.). The size of the pure Fe powder followed the normal distribution and the mean particle size was 49.071 μm . In the simulation, the particle size of 50 μm was adopted for the simplified calculation. The 316 L stainless steel was chosen as the matrix. The properties of the materials were assumed isotropic. All of the physical constants used as put based on the international system of units (SI). Table 1 lists material properties and initial constants for the simulation.

3. Experimental

Laser sintering process was performed by the IPG-50w fibre laser with a CW mode. Single layer sintering with the sample dimension of 6.13 mm mm $\times$ 6.08 mm $\times$ 1.54 mm was executed by a self-developed laser sintering system. The process parameters with the laser power of 5–20 W, scanning speed of 5–30 mm/s, beam diameter of 50 μm and hatching distance of 0.1 mm were adopted in the experiment. The powder bed was not pre-heated before laser sintering. The shielding gas of Ar was adopted during the sintering process to avoid the oxidation of the Fe powder. After laser sintering, the microstructure of sintered samples was observed by optical microscopy (OM) (OLYMPUS U-5RE-2) and scanning electron microscopy (SEM) (HITACHI S-3400N). The

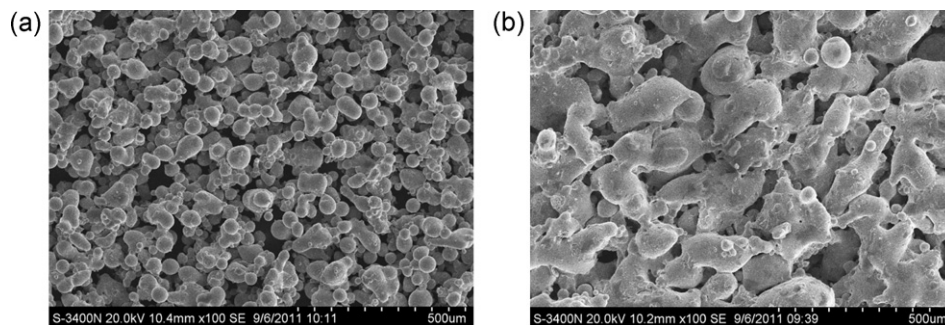


Fig. 3. SEM observations of the samples by selective laser sintering with laser power of $P=5$ W, beam diameter of 50 μm and scanning speed of (a) 20 mm/s and (b) 5 mm/s respectively.

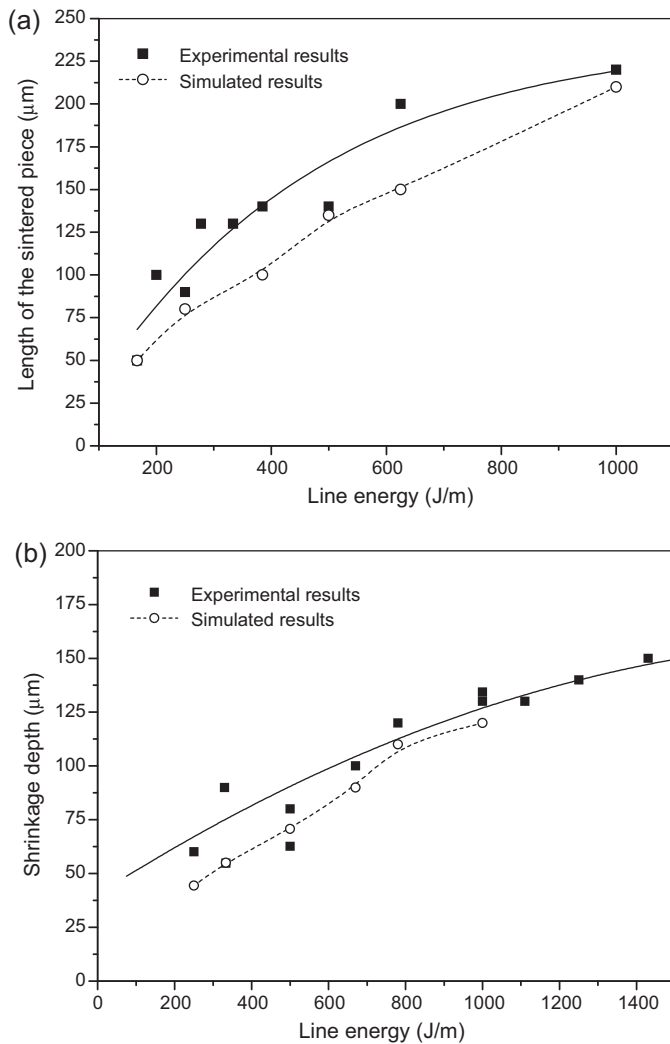


Fig. 4. Comparison between the experimental and simulated results in the (a) length of the sintered piece and (b) shrinkage depth.

length of the sintered piece under each group of process parameters was measured from SEM micrographs. The data was averaged among 10–20 pieces under the same field of view. But the shrinkage depth was not directly measured by the experimental instrument, which was calculated by

$$t_{\text{exp}} = t_0 - t_n - t_s \quad (11)$$

where t_{exp} is shrinkage depth from the experiment, t_0 is initial depth of the powder bed which is set 500 μm in the experiment, t_s is sintered thickness of the sample, and t_n is surplus depth of the powder bed without sintering. t_0 and t_n were measured from the up-down movement of spreading powder pistol with an absolute precision of 1 μm, while t_s was obtained by a micrometer.

4. Results and discussion

4.1. Verification

Fig. 3 compares the SEM observation of sintered samples under different scanning speeds of 20 mm/s and 5 mm/s. It shows that scanning speed has a significant effect on the shape of the sintered piece. For the scanning speed of 20 mm/s, particle melting was poor and the sintered piece held the similar size of the original powder. Also, it is found that sintered powders become near

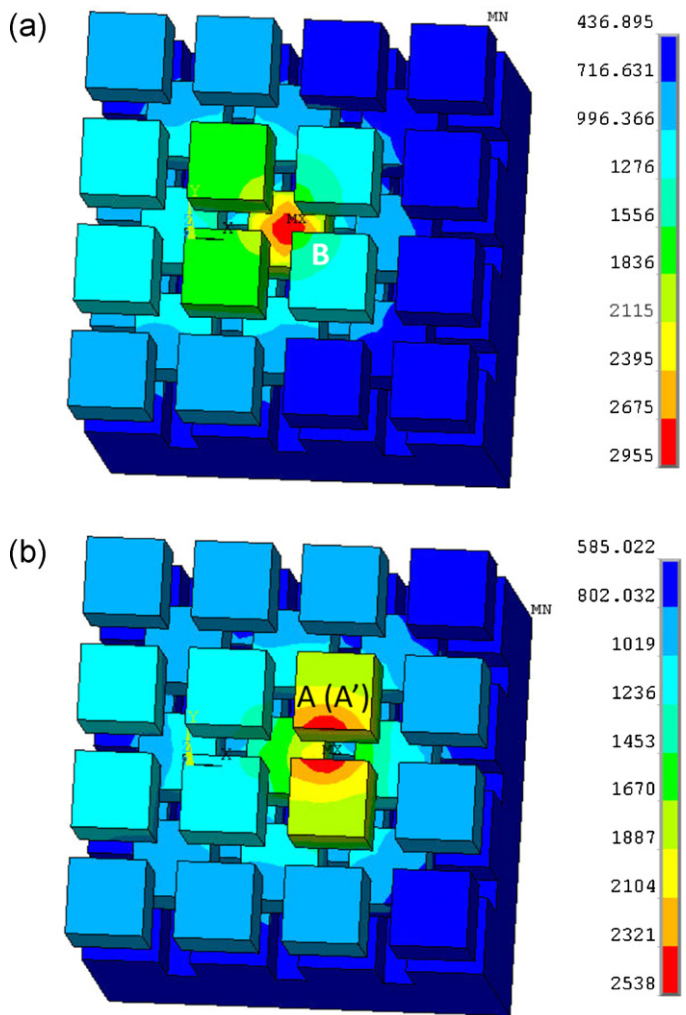


Fig. 5. Temperature profiles at the different times of (a) 8.0 ms and (b) 9.5 ms under case 1.

spherical in shape owing to the melting of the sharp corner as compared with Fig. 2. For the scanning speed of 5 mm/s, as shown in Fig. 3(b), an ellipsoidal piece with the length about 200 μm was observed after laser sintering. The similar work was reported by Fan et al. (2001). They studied the movement of melting pool using in situ video microscopy, and found that powders moved toward the melting pool because of surface tension during laser heating. Therefore, ellipsoidal piece was produced along the laser scanning direction.

The verification of the numerical model is shown in Fig. 4. The simulation results were obtained with the ratio of $R_a = 1.4$ to describe the reasonable porosity of the powder bed. The comparison shows that the simulated length of the sintered piece and shrinkage depth agree well with the experimental data, and both of them increase with the increase of the line energy owing to the bigger melting pool produced by the more energy input. It is also noted that the predicted length of the sintered piece and shrinkage depth are slightly smaller than that from the experiment. The reason for the discrepancies is related to the tension effect as discussed before.

4.2. Thermal field characteristics

In order to clarify the characteristics of the temperature field for the SLS process, the typical process parameters of laser power of 5 W, scanning speed of 20 mm/s, beam diameter of 50 μm as well

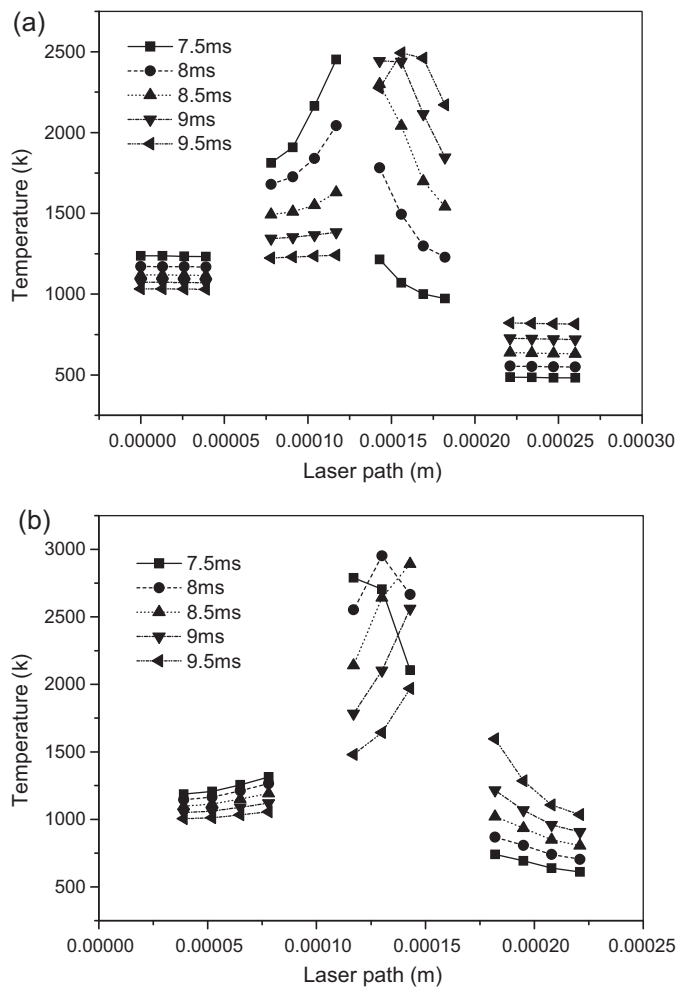


Fig. 6. Temperature distribution along the line (a) MM' and (b) NN' at the different times of 7.5 ms, 8 ms, 8.5 ms, 9 ms and 9.5 ms under case 1.

as space ratio of 1.4 was adopted in the simulation. Fig. 5 shows temperature profiles at different times of 8.0 ms and 9.5 ms under case 1. For the time of 8.0 ms, the maximum temperature is located at the center of the powder in the middle layer. The position of maximum temperature changes to the edge of the powders in the top layer when it is 9.5 ms. A big gap of over 400 K in the maximum temperature between these two situations is obtained. The possible reason comes from the porous structure together with the laser scanning path. Under case 1, because the Gaussian heat flux is distributed along the line L1, which is also along the center of the powder in the middle layer as shown in Fig. 1(c), much more laser power is absorbed by the particles in the middle layer as compared to those in the top layer.

Fig. 6 shows the temperature distribution along the lines of (a) MM' and (b) NN' at different times of 7.5 ms, 8 ms, 8.5 ms, 9 ms and 9.5 ms under case 1. The line MM' is defined along the lower edge of the powders in the second row of the top layer, while the line NN' is along the center of the particles in the second row of the middle layer. The predicted results show that the dwelling time of one particle is about 1 ms, which is in accordance with the previous investigation by Agarwala et al. (1995) in their investigations on laser sintering of metals. The temperature distribution in both layers is not continuous owing to the discretely distributed particles. This intermittent heating pattern will affect the way of laser sintering. It is found that two modes of laser sintering are performed under case 1: one is that laser beam strikes the middle layer of the powder to make them melt and bond the powders around; in

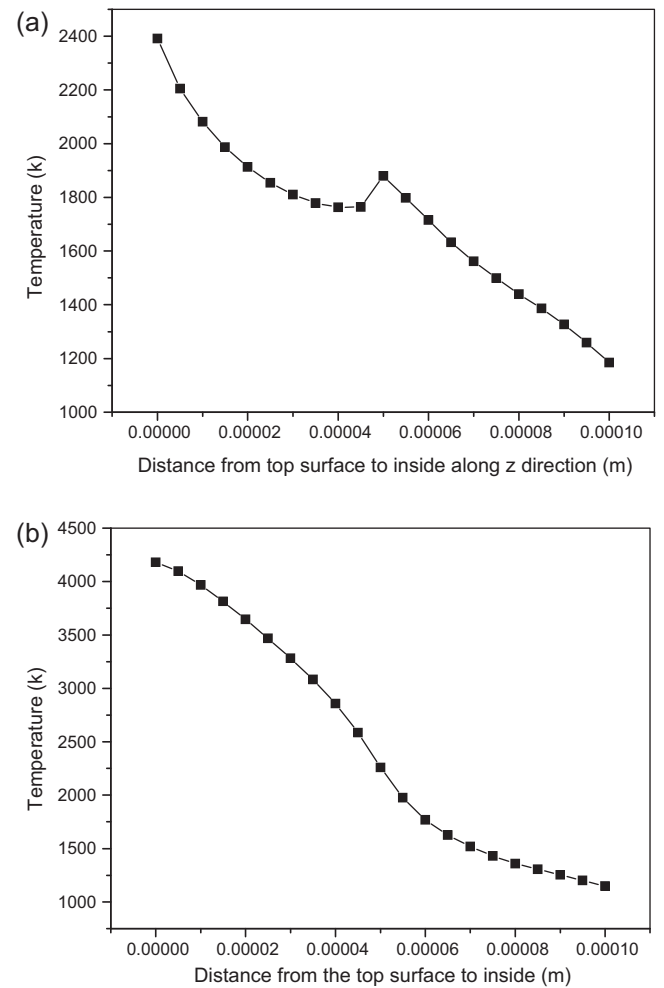


Fig. 7. Temperature variation along z direction under (a) case 1 and (b) case 2.

another way, laser beam melts powders in the top layer and the liquid falls down to adhere the neighboring powders.

Fig. 7(a) shows temperature variation along the line AA' (depth direction), where A and A' shown in Fig. 1(c) stand for two fixed points with the same x and y coordinates. The predicted temperature first decreases quickly within the particle in the top layer, while inversely increases in the region near to the lower edge of the particle owing to heat conduction from the surrounding particles. After then, temperature attenuation shows a linear relationship with the depth in the middle layer.

Fig. 8 gives temperature profiles with different times of 8.0 ms and 9.5 ms under case 2. Different from case 1, the maximum temperature occurs only in the top layer. This is because the center of laser beam is coincident with that of the third row particles so that most laser input power is absorbed by the top layer as compared with case 1. Consequently, for case 2, the way of laser sintering is primarily concluded by falling down of the melting liquid from the top layer to make an adhesion after solidification. Fig. 7(b) shows temperature variation along the line AA' for case 2. Because most of the laser input power is absorbed by the particles in the top layer, the predicted surface temperature is prominently higher than that in case 1, and the trend of attenuation is consistent from the top layer to the middle layer.

Fig. 9 compares temperature variation at the center of one particle under case 1 and case 2. As shown in Figs. 5(a) and 8(b), points of B and C (white circle) are adopted for the simulation respectively. Point B is located in the middle layer for case 1, while point C in the

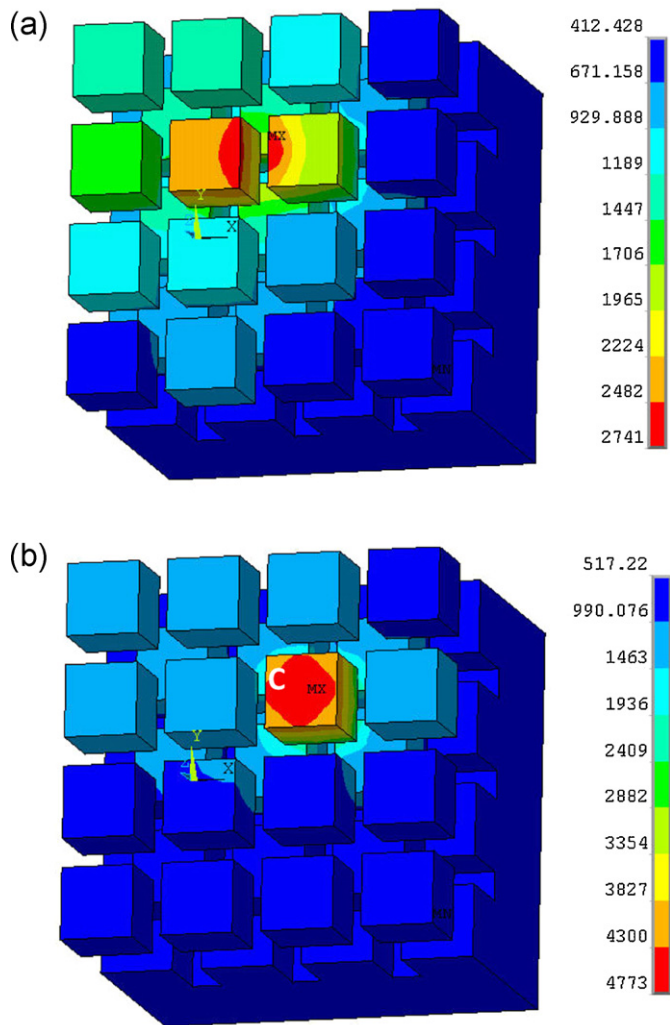


Fig. 8. Temperature profiles at the different times of (a) 8.0 ms and (b) 9.5 ms under case 2.

top layer for case 2. A visible discrepancy in the maximum temperature is found between these two cases. The maximum temperature over boiling point of the material for case 2 means that some parts of powders are vaporized. Also, it is noted that although dwelling

time of the powder is 1 ms under the examined process parameters, the actual melting time of the powder center is about 2.5 ms and 4.8 ms respectively under case 1 and case 2 owing to the heat accumulation effect. In the comparison between case 1 and case 2, it is concluded that laser sintering is significantly inhomogeneous in the micro scale and the maximum temperature of the powder bed occurs at the center of the powder in the top layer under case 2.

5. Conclusions

A 3D micro scale finite element model was built up to investigate thermal evolution of the porous powder bed in selective laser sintering of metal powders. The significant founding is summarized as follows:

- (1) The simulated length of the sintered piece and shrinkage depth of the powder bed showed a reasonable agreement with the experiment data. Both of them increased with the increase of the line energy owing to the bigger melting pool produced by the more energy input.
- (2) The typical temperature field characteristics of the SLS process were studied under two cases of laser scanning strategies (case 1 and case 2). It showed that temperature field of laser sintering was intermittent in the micro scale due to the discretely distributed particles and temperature distribution was significantly inhomogeneous with maximum temperature of the powder bed produced in the top layer under case 2.
- (3) For the mechanism of laser sintering, two kinds of performance were executed in the powder bed. One was that the laser beam struck the middle layer of the powder to make them melt and bond the powders around; the other was by the falling down of the melting liquid from the top layer.

Acknowledgments

The work described in this paper was supported by research findings from Beijing Municipal (no:2009D005015000007) and Beijing University of Technology (no: X4101011201101).

References

- Agarwala, M., Bourell, D., Beaman, J., Marcus, H., Barlow, J., 1995. Direct selective laser sintering of metals. *Rapid Prototyping Journal* 1 (1), 26–36.
- Anestiev, L.A., Froyen, L., 1999. Model of the primary rearrangement processes at liquid phase sintering and selective laser sintering due to biparticle interactions. *Journal of Applied Physics* 86 (7), 4008–4017.
- Bugeda, G., Cervera, M., Lombera, G., 1999. Numerical prediction of temperature and density distributions in selective laser sintering processes. *Rapid Prototyping Journal* 5, 21–26.
- Chen, T.B., Zhang, Y.W., 2007. Thermal modeling of laser sintering of two-component metal powder on top of sintered layer via multi-line scanning. *Applied Physics A* 84, 213–220.
- Dai, K., Shaw, L., 2004. Thermal and mechanical finite element modeling of laser forming from metal and ceramic powders. *Acta Materialia* 52, 69–80.
- Dai, K., Shaw, L., 2005. Finite element analysis of the effect of volume shrinkage during laser densification. *Acta Materialia* 53, 4743–4754.
- Deckard, C.R., 1986. Part generation by layer-wise selective laser sintering. M. Sc. Thesis. The University of Texas at Austin, Austin, USA.
- Dong, L., Makrady, A., Ahzi, S., Remond, Y., 2009. Three-dimensional transient finite element analysis of the selective laser sintering process. *Journal of Materials Processing Technology* 209, 700–706.
- Fan, K.M., Cheung, W.L., Gibson, I., 2001. Material movement and fusion behavior of true form/SiO₂ during selective laser sintering. In: Bourell, D.L., et al. (Eds.), *Proceedings of the 12th Annual SFF Symposium*. Austin, pp. 146–154.
- Gusarov, A.V., Laoui, T., et al., 2003. Contact thermal conductivity of a powder bed in selective laser sintering. *International Journal of Heat and Mass Transfer* 46, 1103–1109.
- Kolossov, S., Boillat, E., Gardon, R., Fischer, P., Locher, M., 2004. 3D FE simulation for temperature evolution in the selective laser sintering process. *International Journal of Machine Tools and Manufacture* 44, 117–123.

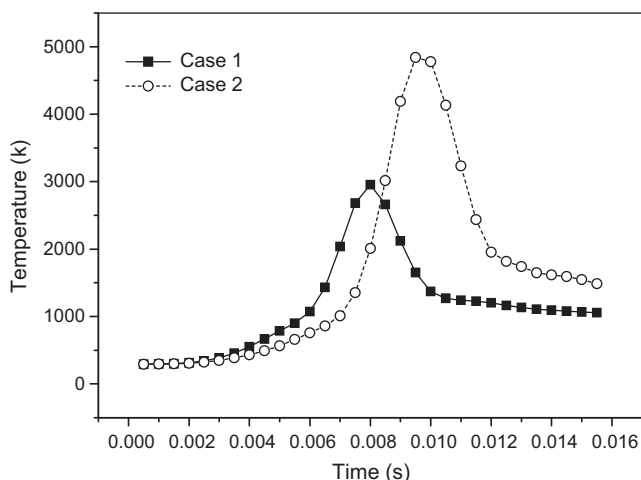


Fig. 9. Comparison of the temperature variation at the center of the powder between case 1 and case 2.

- Nelson, J.C., Xue, S., Barlow, J.W., Beaman, J.J., Marcus, H.L., Bourell, D.L., 1993. Model of the selective laser sintering of bisphenol-a polycarbonate. *Industrial Engineering and Chemical Research* 32, 2305–2317.
- Salmoria, G.V., Klauss, P.K., Paggi, R.A., Kanis, L.A., Lago, A., 2009. Structure and mechanical properties of cellulose based scaffolds fabricated by selective laser sintering. *Polymer Test* 28, 648–652.
- Song, J.L., Li, Y.T., Deng, Q.L., Hu, D.J., 2007. Rapid prototyping manufacturing of silica sand patterns based on selective laser sintering. *Journal of Materials Processing Technology* 187–188, 614–618.
- Tolochko, N.K., Laoui, T., Khlopkov, Y.V., Mozzharov, S.E., Titov, V.I., Ignatiev, M.B., 2000. Absorbance of powder materials suitable for laser sintering. *Rapid Prototyping Journal* 6 (3), 155–160.
- Williams, J.D., Deckard, C.R., 1998. Advances in modeling the effects of selected parameters on the SLS process. *Rapid Prototyping Journal* 4, 90–100.
- Yan, C.H., Hao, L., Xu, L., Shi, Y., 2011. Preparation, characterization and processing of carbon fibre/polyamide-12 composites for selective laser sintering. *Composites Science and Technology* 71, 1834–1841.
- Zhang, Y., Faghri, A., 1999. Melting of a subcooled mixed powder bed with constant heat flux heating. *International Journal of Heat and Mass Transfer* 42, 775–788.